Alpha Diversity Analysis

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Contents

Alpha Diversity of mice based on age, sex, and cohort

Based on the below results, I determined that sex difference in alpha diversity were minimal. The alpha diversity and age do not have a linear relationship. Increase in diversity was observed after 5 months old, with a moderate drop after 19 months. The effect was more in "Observed" diversity index, more so than Shannon.

```r
## alpha diversity ====
library(lme4)
```

```
## Loading required package: Matrix
```

```r
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##     lmer

## The following object is masked from 'package:stats':
##
##     step
```

```r
library(knitr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(phyloseq)
library(vegan)
```

```
## Loading required package: permute
```

```r
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(tibble)
library(reshape2)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.2     v stringr   1.5.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(qiime2R)
library(ggpubr)
```

```
##
## Attaching package: 'ggpubr'
##
## The following object is masked from 'package:qiime2R':
##
##     mean_sd
```

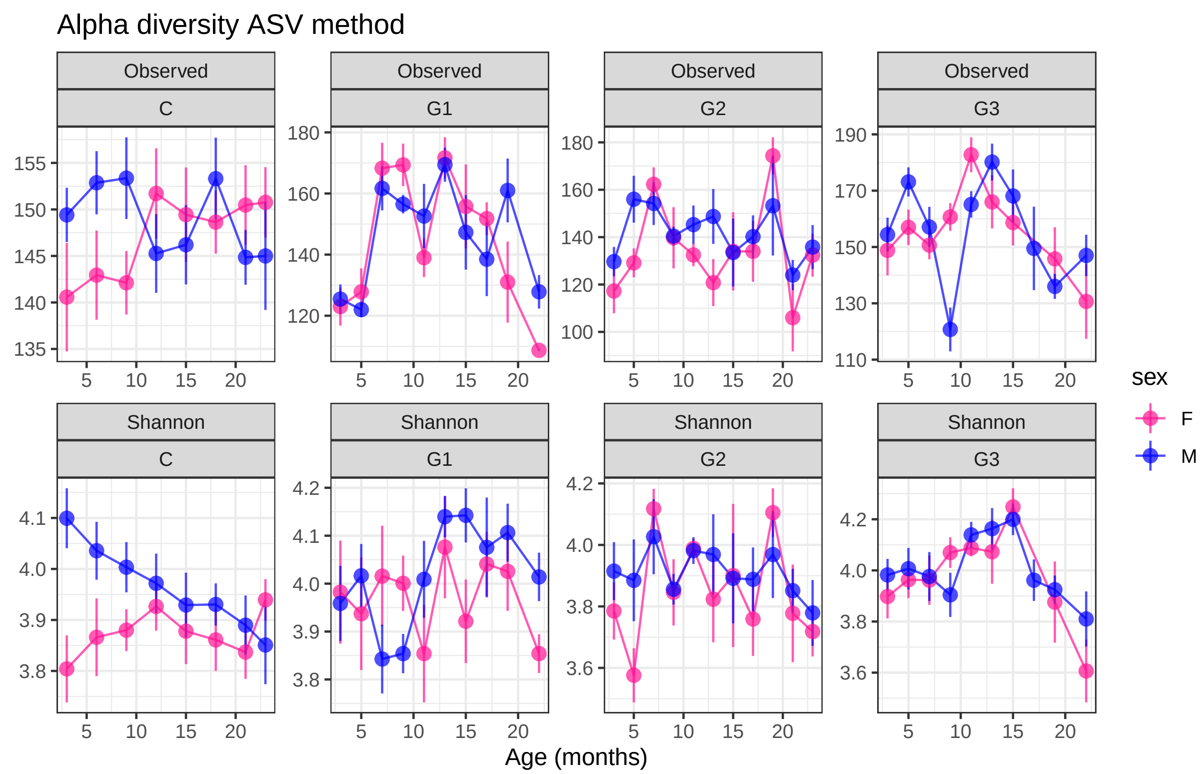
```r
library(ggplot2)

theme_set(theme_bw())

load("qiime_f.RData")
metadata_df<-as.data.frame(sample_data(qiime_f))
metadata_df$SampleID<-rownames(metadata_df)
metadata_df<-as.matrix(metadata_df)

qiime_f_rare<-rarefy_even_depth(qiime_f,sample.size = min(sample_sums(qiime_f)))
```

```
## You set `rngseed` to FALSE. Make sure you've set & recorded
##  the random seed of your session for reproducibility.
## See `?set.seed`
##
## ...
## 1662OTUs were removed because they are no longer
## present in any sample after random subsampling
##
## ...
```

```r
alpha.div.df <- estimate_richness(
  qiime_f_rare,
  measures=c("Shannon", "Observed")) %>%
  rownames_to_column("SampleID") %>%
  merge(metadata_df, by="SampleID")
View(alpha.div.df)
alpha.div.long.df <- alpha.div.df %>%
  pivot_longer(c(Shannon, Observed), names_to="metric")

alpha.div.long.df$month_n<-as.numeric(alpha.div.long.df$month_n)

# summarize across sexes
alpha.div.plot.df <- alpha.div.long.df %>%
  dplyr::filter(metric %in% c("Shannon", "Observed")) %>%
  dplyr::filter(month_n <= 40) %>% # omit the few 46 month samples

  # compute mean and SD across diets (separately for each metric)
  group_by(metric, sex, month_n, mousegroup.x) %>%
  summarise(n=n(),
            mean = mean(value),
            sd = sd(value),
            sem = sd(value)/sqrt(n()), .groups="drop") %>%

  # make sure we have at least 3 observations at this age combination
  dplyr::filter(n >= 2) #%>%

#plot Fig 2 alpha ====
alpha.div.plot.df$month_n <- as.numeric(alpha.div.plot.df$month_n)
alpha.div.plot.df <- alpha.div.plot.df %>%
  arrange(metric, sex, month_n)
```

Alpha diversity plot

```
alpha.div.plot.df %>%
  ggplot(aes(
    x = month_n, y = mean,
    ymin = mean - sem, ymax = mean + sem,
    color = sex, group = interaction(sex, metric)
  )) +
  geom_path(alpha = 0.7) +
  geom_pointrange(alpha = 0.7) +
  facet_wrap(~ metric + mousegroup.x, scales = "free", nrow = 2) +
  scale_color_manual(values = c("deeppink", "blue")) +
  labs(
    x = "Age (months)",
    y = "",
    title = "Alpha diversity ASV method"
  ) +
  theme_bw()
```



LMER for alpha

```
df_wide <- alpha.div.long.df %>%
  pivot_wider(names_from = metric, values_from = value)
```

Q1: Are there differences between sex and month, and if so, when?

Q2: Which months differ from baseline in males and females separately?

```
model1 <- lmer(Shannon ~ sex * month + (1 | mousegroup.x) + (1 | mouse), data = df_wide)
kable(anova(model1, type = 3), caption = "ANOVA for model1 with all cohort")
```

Table 1: ANOVA for model1 with all cohort

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	0.561285	0.5612850	1	103.3558	10.518604	0.0015918
month	1.801652	0.1286894	14	467.9999	2.411668	0.0028663
sex:month	1.122979	0.0802128	14	561.0674	1.503205	0.1047599

```
kable(summary(model1)$coefficients, digits = 4)
```

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.8563	0.0491	34.7274	78.5894	0.0000
sexM	0.1521	0.0553	590.0081	2.7525	0.0061
monthM05	-0.0497	0.0702	554.1265	-0.7080	0.4792
monthM06	0.0157	0.0731	551.1971	0.2144	0.8303
monthM07	0.1771	0.0680	551.7799	2.6061	0.0094
monthM09	0.0645	0.0574	549.9732	1.1238	0.2616
monthM11	0.0699	0.0716	563.2445	0.9762	0.3294
monthM12	0.0759	0.0764	558.4325	0.9936	0.3209
monthM13	0.1342	0.0744	562.0013	1.8052	0.0716
monthM15	0.0686	0.0598	543.0441	1.1456	0.2525
monthM17	0.0890	0.0731	562.3241	1.2184	0.2236
monthM18	0.0121	0.0829	556.6897	0.1458	0.8841
monthM19	0.1535	0.0715	560.3498	2.1490	0.0321
monthM21	-0.0140	0.0699	551.1546	-0.2010	0.8408
monthM22	-0.1998	0.0897	568.8409	-2.2274	0.0263
monthM23	0.0515	0.0657	548.0827	0.7837	0.4335
sexM:monthM05	0.0106	0.0926	561.8637	0.1144	0.9089
sexM:monthM06	0.0143	0.0999	570.7765	0.1431	0.8863
sexM:monthM07	-0.2599	0.0892	559.1109	-2.9145	0.0037
sexM:monthM09	-0.1567	0.0782	547.3468	-2.0044	0.0455
sexM:monthM11	-0.0303	0.0914	564.0387	-0.3319	0.7401
sexM:monthM12	-0.1100	0.1024	571.5838	-1.0744	0.2831
sexM:monthM13	-0.0520	0.0964	565.5612	-0.5390	0.5901
sexM:monthM15	-0.0490	0.0789	541.1258	-0.6219	0.5343
sexM:monthM17	-0.1230	0.0980	569.5130	-1.2551	0.2100
sexM:monthM18	-0.0917	0.1099	572.2955	-0.8338	0.4048
sexM:monthM19	-0.1510	0.0968	566.0482	-1.5595	0.1194
sexM:monthM21	-0.0929	0.0948	564.5730	-0.9798	0.3276
sexM:monthM22	0.0722	0.1165	576.0326	0.6192	0.5360
sexM:monthM23	-0.2094	0.0882	557.8815	-2.3730	0.0180

Interpretation of the LMM summary

In the above, F M03 is reference. sexM result suggests at M03, Male sig. higher than female at M03 (by 0.1515). Females at M07, M19, N22 were sig different from M03. The sex difference in M07(p=.004), M09 (p=0.03), M23(p=0.0122) is significantly different from the sex difference in M03 .

ANOVA for model1 (Shannon with all cohorts)

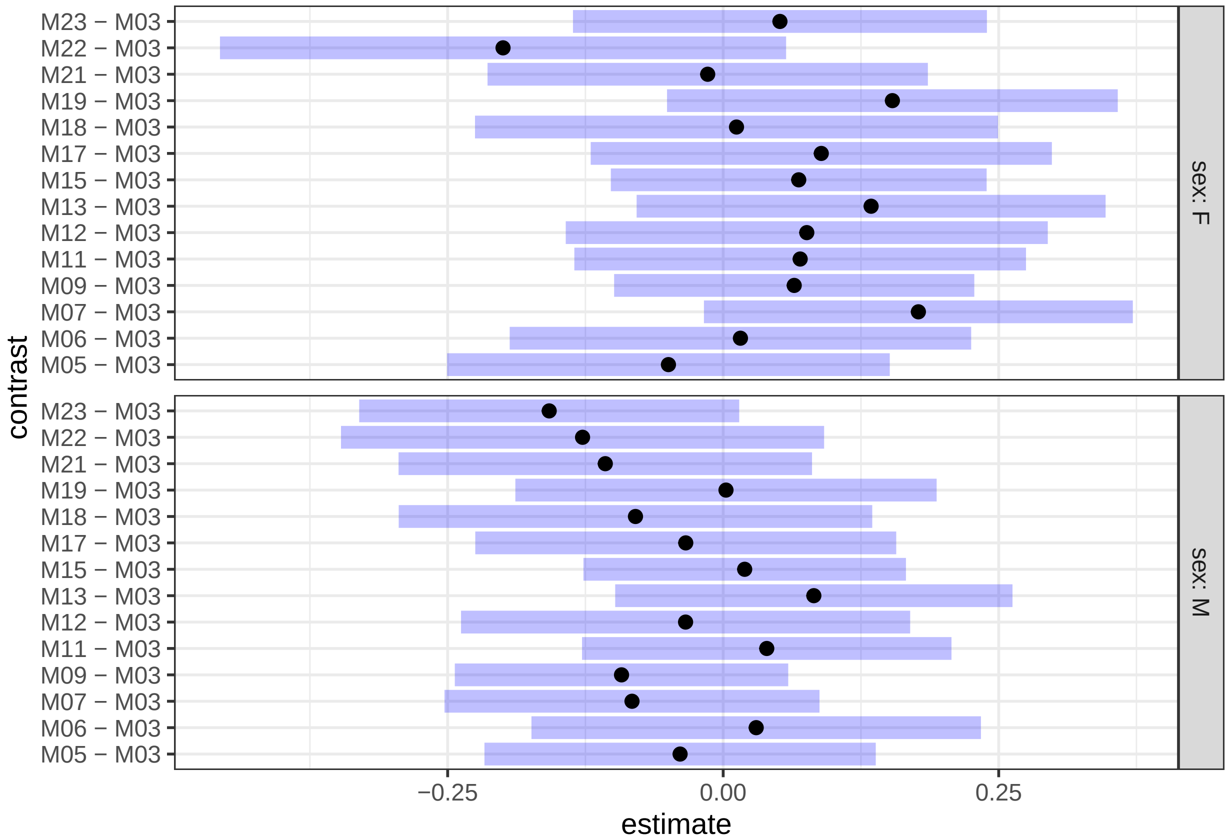
	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	0.5794963	0.5794963	1	102.2983	10.870027	0.0013445
month	1.8934174	0.1352441	14	474.3694	2.536870	0.0016396
sex:month	1.2073909	0.0862422	14	559.5593	1.617707	0.0700462

```
# Estimated marginal means by Sex within Month for Shannon
emm_sex_within_month <- emmeans(model1, ~ sex | month)
pairs_sex_within_month <- contrast(emm_sex_within_month, method = "pairwise", adjust = "holm")
pairs_sex_within_month
```

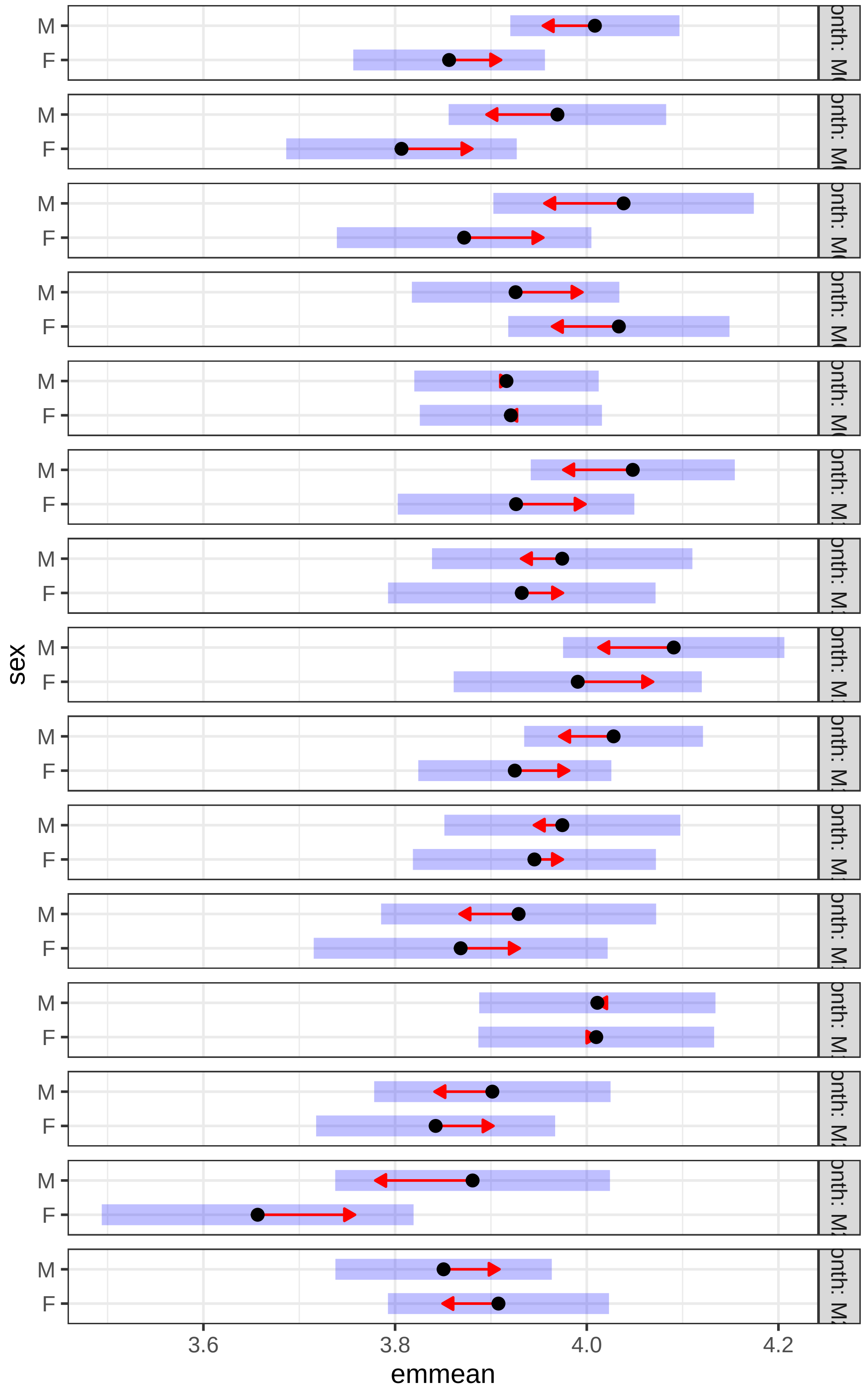
```
## month = M03:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.15211 0.0553 590  -2.750  0.0061
##
## month = M05:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.16270 0.0754 591  -2.158  0.0313
##
## month = M06:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.16641 0.0842 591  -1.975  0.0487
##
## month = M07:
##  contrast estimate     SE  df t.ratio p.value
##  F - M     0.10776 0.0711 591   1.516  0.1301
##
## month = M09:
##  contrast estimate     SE  df t.ratio p.value
##  F - M     0.00458 0.0568 591   0.081  0.9358
##
## month = M11:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.12179 0.0739 592  -1.649  0.0998
##
## month = M12:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.04215 0.0871 591  -0.484  0.6287
##
## month = M13:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.10013 0.0801 591  -1.250  0.2116
##
## month = M15:
```

```
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.10306 0.0578 591  -1.783  0.0751
##
## month = M17:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.02914 0.0820 593  -0.355  0.7225
##
## month = M18:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.06044 0.0960 591  -0.629  0.5294
##
## month = M19:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.00112 0.0805 591  -0.014  0.9889
##
## month = M21:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.05919 0.0782 591  -0.757  0.4495
##
## month = M22:
##  contrast estimate     SE  df t.ratio p.value
##  F - M    -0.22426 0.1030 591  -2.167  0.0306
##
## month = M23:
##  contrast estimate     SE  df t.ratio p.value
##  F - M     0.05726 0.0699 590   0.819  0.4131
##
## Degrees-of-freedom method: kenward-roger
```

```r
emm_month_within_sex <- emmeans(model1, ~ month | sex)
baseline_contrasts <- contrast(emm_month_within_sex, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts)
```




```
plot(emm_sex_within_month, comparisons = TRUE)
```



Interpretation of EMMeans figure

Within each month, we are looking at sex differences. Overall, the females have a higher alpha diversity than males.

Subsetting per cohort

Cohort G1

```
df_wide_G1<- df_wide %>%
  filter(mousegroup.x== "G1")
View(df_wide_G1)

model1_G1 <- lmer(Shannon ~ sex * month + (1 | mouse), data = df_wide_G1)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(model1_G1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Shannon ~ sex * month + (1 | mouse)
##    Data: df_wide_G1
##
## REML criterion at convergence: 12.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9498 -0.5371  0.1819  0.6750  1.8534
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  mouse    (Intercept) 0.00000  0.0000
##  Residual             0.04742  0.2178
## Number of obs: 142, groups:  mouse, 19
##
## Fixed effects:
##                 Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)      3.98221    0.08231 122.00000  48.381   <2e-16 ***
## sexM            -0.02350    0.10974 122.00000  -0.214    0.831
## monthM05        -0.04509    0.12115 122.00000  -0.372    0.710
## monthM07         0.03347    0.11640 122.00000   0.288    0.774
## monthM09         0.01849    0.11271 122.00000   0.164    0.870
## monthM11        -0.12828    0.10974 122.00000  -1.169    0.245
## monthM13         0.09395    0.12115 122.00000   0.775    0.440
## monthM15        -0.06078    0.11271 122.00000  -0.539    0.591
## monthM17         0.05860    0.10974 122.00000   0.534    0.594
## monthM19         0.04341    0.12115 122.00000   0.358    0.721
## monthM22        -0.12816    0.15027 122.00000  -0.853    0.395
## sexM:monthM05    0.10279    0.17156 122.00000   0.599    0.550
## sexM:monthM07   -0.14955    0.15520 122.00000  -0.964    0.337
## sexM:monthM09   -0.12320    0.15459 122.00000  -0.797    0.427
## sexM:monthM11    0.17848    0.15245 122.00000   1.171    0.244
## sexM:monthM13    0.08707    0.16347 122.00000   0.533    0.595
## sexM:monthM15    0.24455    0.15731 122.00000   1.555    0.123
## sexM:monthM17    0.05828    0.15880 122.00000   0.367    0.714
## sexM:monthM19    0.10434    0.16086 122.00000   0.649    0.518
```

```
## sexM:monthM22   0.18349    0.18909 122.00000   0.970    0.334
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 20 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# From the summary result: There was no difference between sexes at M03, or difference with females, or
```

```
kable(anova(model1_G1, type = 3), caption ="ANOVA for Cohort G1: Shannon")
```

Table 4: ANOVA for Cohort G1: Shannon

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	0.0676959	0.0676959	1	122	1.427497	0.2344908
month	0.5637940	0.0626438	9	122	1.320964	0.2328577
sex:month	0.5525425	0.0613936	9	122	1.294601	0.2465863

```
model2_G1 <- lmer(Observed ~ sex * month + (1 | mouse), data = df_wide_G1)
```

```
## boundary (singular) fit: see help('isSingular')
```

cohort G1: Observed

From the below table:

1. There is no difference between male and females at M03
2. Females at month M07, M09, M13, M16, and M17 are significantly different from females at M03
3. Compared to the difference between males and females in M03, there was no sig difference between Males and Females in other months

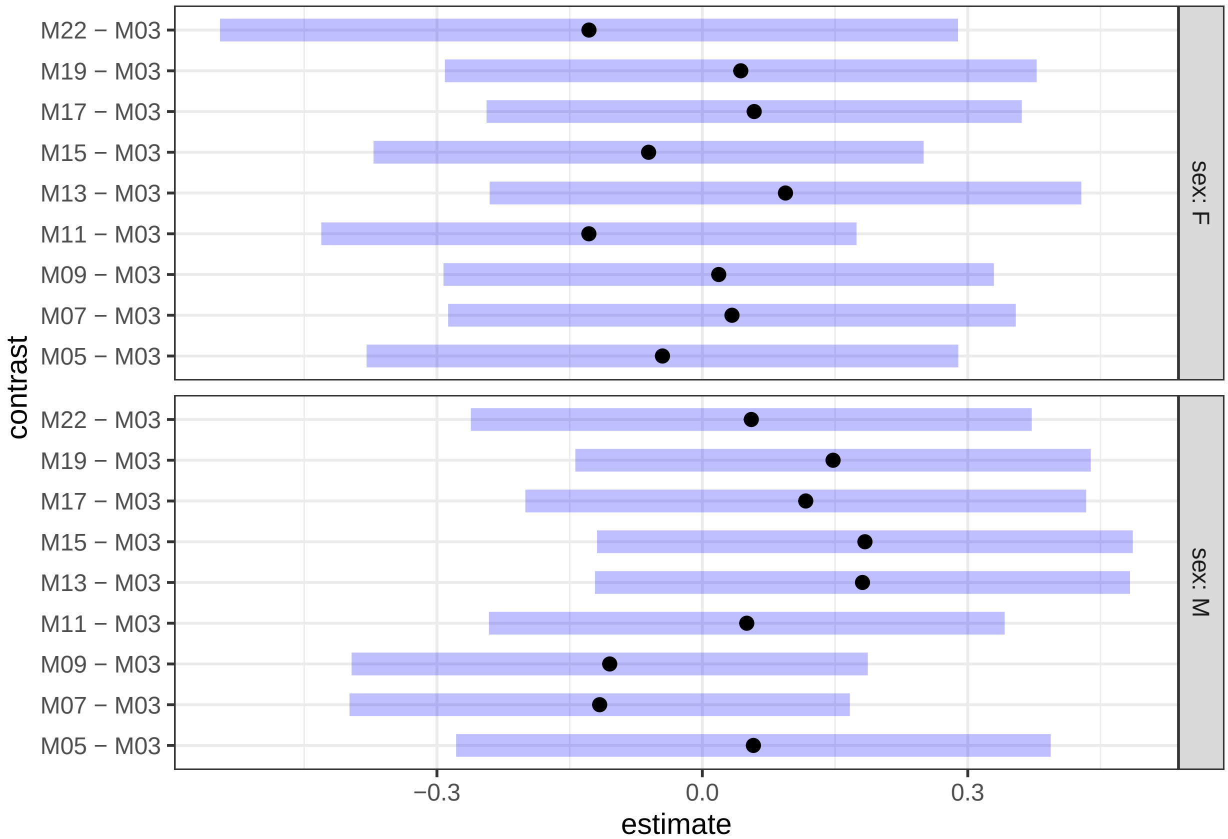
	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	123.1429	8.4062	122	14.6491	0.0000
sexM	2.3016	11.2082	122	0.2053	0.8376
monthM05	5.5238	12.3736	122	0.4464	0.6561
monthM07	45.4286	11.8881	122	3.8213	0.0002
monthM09	45.8571	11.5106	122	3.9839	0.0001
monthM11	16.9683	11.2082	122	1.5139	0.1326
monthM13	46.1905	12.3736	122	3.7330	0.0003
monthM15	32.1071	11.5106	122	2.7893	0.0061
monthM17	28.6349	11.2082	122	2.5548	0.0119
monthM19	7.0238	12.3736	122	0.5676	0.5713
monthM22	-14.4762	15.3475	122	-0.9432	0.3474

	Estimate	Std. Error	df	t value	Pr(>|t|)
sexM:monthM05	-8.7683	17.5213	122	-0.5004	0.6177
sexM:monthM07	-6.4286	15.8509	122	-0.4056	0.6858
sexM:monthM09	-15.0516	15.7888	122	-0.9533	0.3423
sexM:monthM11	8.2123	15.5697	122	0.5275	0.5988
sexM:monthM13	-2.4921	16.6952	122	-0.1493	0.8816
sexM:monthM15	-12.5516	16.0661	122	-0.7812	0.4362
sexM:monthM17	-15.0794	16.2181	122	-0.9298	0.3543
sexM:monthM19	28.5317	16.4285	122	1.7367	0.0850
sexM:monthM22	17.1984	19.3119	122	0.8906	0.3749

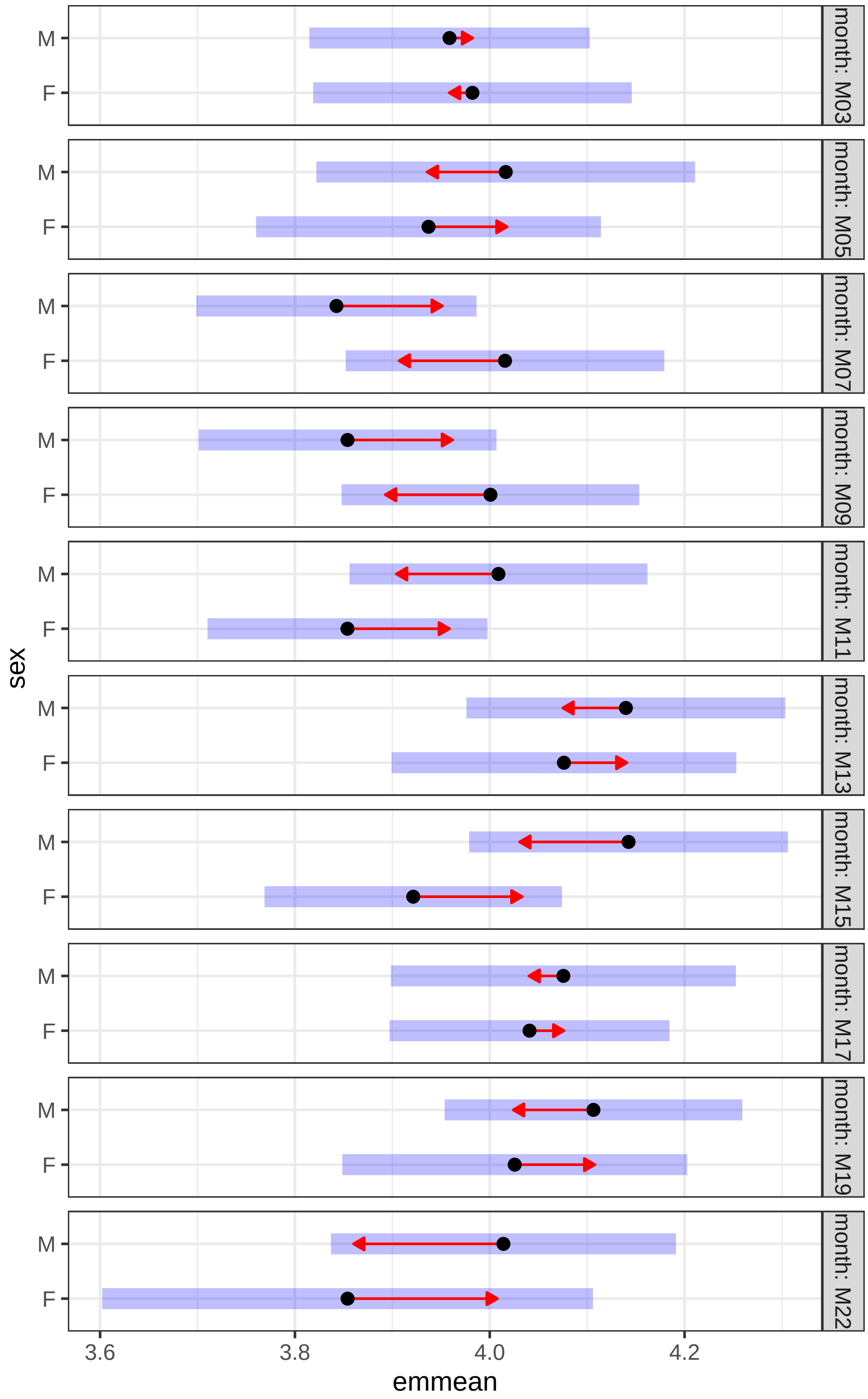
ANOVA of ModelG1: Observed

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	91.47856	91.47856	1	122	0.1849369	0.6679231
month	36277.26342	4030.80705	9	122	8.1488478	2.09e-09 ***
sex:month	6297.16741	699.68527	9	122	1.4145130	0.1890481

```r
# Estimated marginal means by Sex within Month for Shannon
emm_sex_within_month_G1 <- emmeans(model1_G1, ~ sex | month)
pairs_sex_within_month_G1 <- contrast(emm_sex_within_month, method = "pairwise", adjust = "holm")

emm_month_within_sex_G1 <- emmeans(model1_G1, ~ month | sex)
baseline_contrasts_G1 <- contrast(emm_month_within_sex_G1, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_G1)
```




```
plot(emm_sex_within_month_G1, comparisons = TRUE)
```



month: M03
month: M05
month: M07
month: M09
month: M11
month: M13
month: M15
month: M17
month: M19
month: M22
M
F
sex
3.6
3.8
4.0
4.2
emmean

Cohort C

```r
df_wide_C<- df_wide %>%
  filter(mousegroup.x== "C")
model1_C <- lmer(Shannon ~ sex * month + (1 | mouse), data = df_wide_C)
```

Summary Cohort C: Shannon

From the below table:

1. There is sig. difference between male and females at M03 (p=0.0005)
2. Females shannon diversity relatively stable with age
3. Compared to the difference between males and females in M03, there was sig difference (decreasing) between Males and Females in months M12 (p= 0.0347), M15 (p=0.0320), and M23 (p=0.0006)

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.8023	0.0608	219.0609	62.5293	0.0000
sexM	0.2902	0.0821	218.7018	3.5329	0.0005
monthM06	0.0473	0.0810	193.8009	0.5838	0.5600
monthM09	0.0858	0.0792	201.6122	1.0829	0.2802
monthM12	0.1241	0.0838	196.9755	1.4807	0.1403
monthM15	0.0797	0.0810	193.8009	0.9837	0.3265
monthM18	0.0662	0.0891	193.1763	0.7423	0.4588
monthM21	0.0266	0.0822	192.6006	0.3238	0.7464
monthM23	0.1353	0.0798	192.4630	1.6951	0.0917
sexM:monthM06	-0.1082	0.1128	195.5182	-0.9591	0.3387
sexM:monthM09	-0.1615	0.1170	202.4634	-1.3803	0.1690
sexM:monthM12	-0.2441	0.1148	196.0534	-2.1268	0.0347
sexM:monthM15	-0.2437	0.1128	195.9673	-2.1600	0.0320
sexM:monthM18	-0.2234	0.1210	196.4263	-1.8459	0.0664
sexM:monthM21	-0.2252	0.1161	198.1406	-1.9398	0.0538
sexM:monthM23	-0.3868	0.1101	193.5140	-3.5121	0.0006

```r
kable(anova(model1_C, type = 3), caption = "ANOVA for model1 (Shannon)_C")
```

Table 8: ANOVA for model1 (Shannon)_C

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	0.3597627	0.3597627	1	36.31618	7.3249270	0.0102970
month	0.2248599	0.0321228	7	197.62000	0.6540352	0.7107067
sex:month	0.6941729	0.0991676	7	197.62000	2.0190951	0.0544360

```r
model2_C <- lmer(Observed ~ sex * month + (1 | mouse), data = df_wide_C)
```

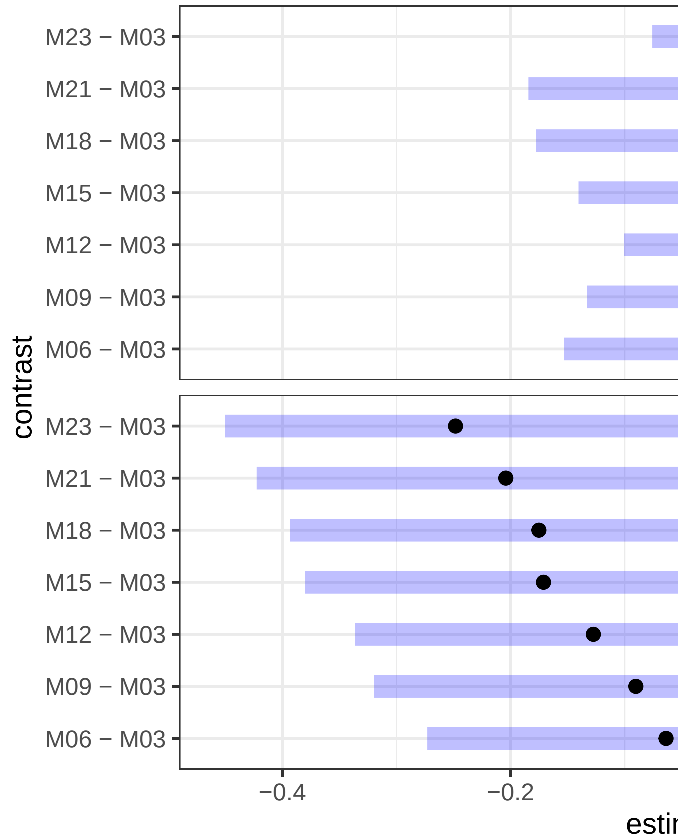
```
## boundary (singular) fit: see help('isSingular')
```

```r
kable(summary(model2_C)$coefficients, digits = 4, caption = "Fixed effects for Observed model (Cohort C
```

Table 9: Fixed effects for Observed model (Cohort C)

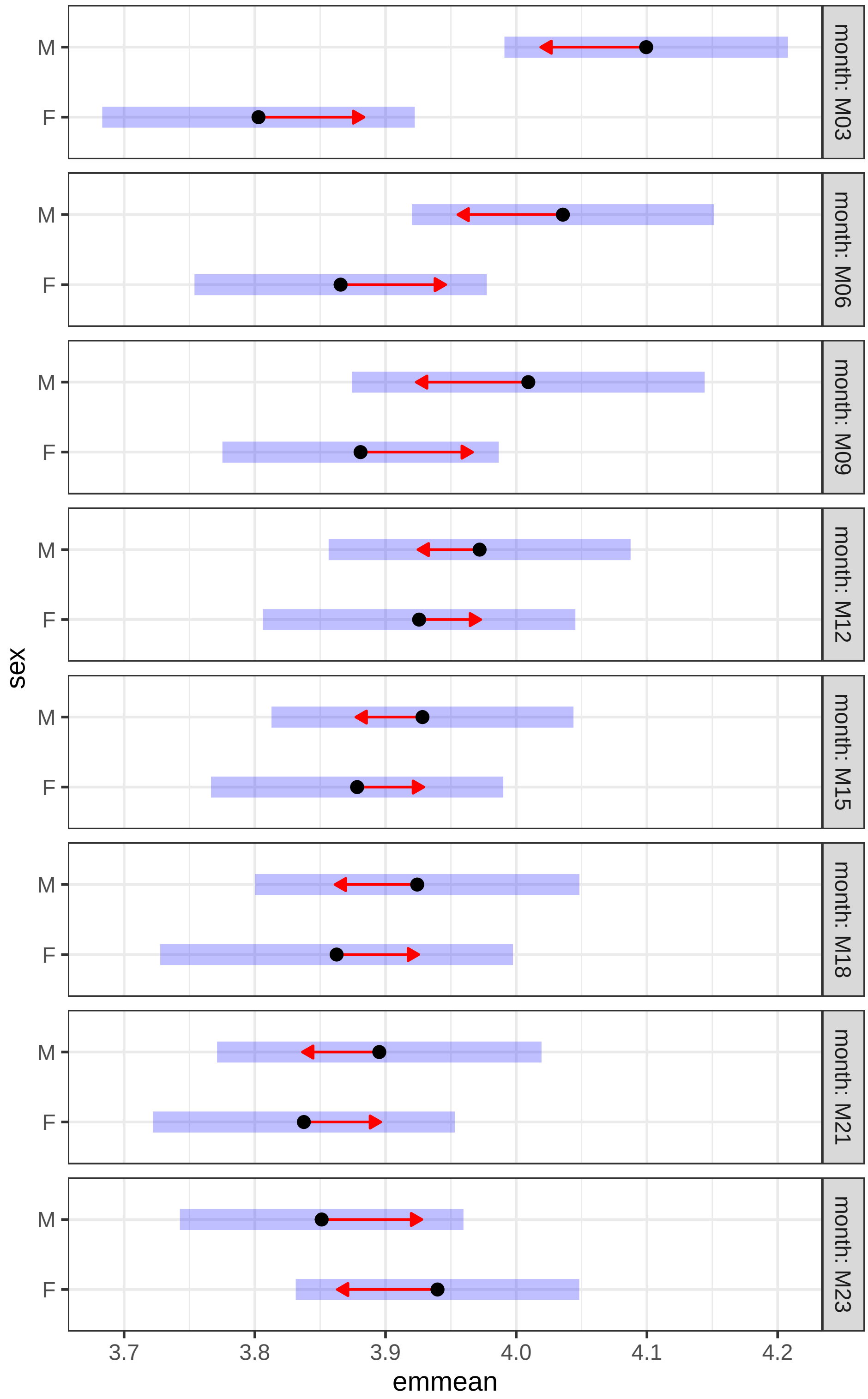
	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	140.5714	4.5032	221	31.2159	0.0000
sexM	8.8403	6.0810	221	1.4538	0.1474
monthM06	2.3661	6.1663	221	0.3837	0.7016
monthM09	1.5397	6.0043	221	0.2564	0.7979
monthM12	11.1429	6.3685	221	1.7497	0.0816
monthM15	8.8661	6.1663	221	1.4378	0.1519
monthM18	8.0649	6.7888	221	1.1880	0.2361
monthM21	9.8952	6.2614	221	1.5803	0.1155
monthM23	10.1933	6.0810	221	1.6762	0.0951
sexM:monthM06	1.0888	8.5819	221	0.1269	0.8992
sexM:monthM09	2.4122	8.8634	221	0.2722	0.7858
sexM:monthM12	-15.2880	8.7284	221	-1.7515	0.0812
sexM:monthM15	-12.0778	8.5819	221	-1.4074	0.1607
sexM:monthM18	-4.1690	9.1993	221	-0.4532	0.6509
sexM:monthM21	-14.4608	8.8173	221	-1.6401	0.1024
sexM:monthM23	-14.6050	8.3892	221	-1.7409	0.0831

```r
# Estimated marginal means by Sex within Month for Shannon
emm_sex_within_month_C <- emmeans(model1_C, ~ sex | month)
pairs_sex_within_month_C <- contrast(emm_sex_within_month_C, method = "pairwise", adjust = "holm")

emm_month_within_sex_C <- emmeans(model1_C, ~ month | sex)
baseline_contrasts_C <- contrast(emm_month_within_sex_C, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_C)
```



In cohort C, no differences in Observed diversity metric.

```
plot(emm_sex_within_month_C, comparisons = TRUE)
```



For Cohort G2

SHANNON

1. Sig. diff between M and F in M03
2. Females are stable
3. No differences in M and F

Table 10: Fixed effects for Observed model (Cohort G2)

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.7939	0.1246	101.2543	30.4592	0.0000
sexM	0.1223	0.1505	99.3598	0.8126	0.4184
monthM05	-0.2431	0.1490	92.9471	-1.6313	0.1062
monthM07	0.3279	0.1489	92.7019	2.2017	0.0302
monthM09	0.0667	0.1499	94.2028	0.4450	0.6574
monthM11	0.1733	0.1713	97.3980	1.0116	0.3142
monthM13	0.0261	0.1600	94.1260	0.1631	0.8708
monthM15	0.1091	0.1847	96.9481	0.5908	0.5560
monthM17	-0.0433	0.1544	94.8884	-0.2803	0.7799
monthM19	0.3025	0.1544	94.8884	1.9587	0.0531
monthM21	-0.0113	0.1825	94.0437	-0.0618	0.9509
monthM23	-0.0786	0.1600	94.2136	-0.4913	0.6244
sexM:monthM05	0.2209	0.1907	92.3040	1.1579	0.2499
sexM:monthM07	-0.2129	0.1995	92.9569	-1.0674	0.2886
sexM:monthM09	-0.1397	0.1915	93.3516	-0.7293	0.4677
sexM:monthM11	-0.1075	0.2087	95.8024	-0.5149	0.6078
sexM:monthM13	0.0446	0.1995	93.2802	0.2236	0.8236
sexM:monthM15	-0.1136	0.2231	95.7534	-0.5091	0.6119
sexM:monthM17	0.0029	0.2036	94.1650	0.0142	0.9887
sexM:monthM19	-0.2752	0.2108	94.4461	-1.3056	0.1949
sexM:monthM21	-0.0380	0.2256	93.7753	-0.1683	0.8667
sexM:monthM23	-0.0581	0.2079	93.9131	-0.2794	0.7805

OBSERVED For G2

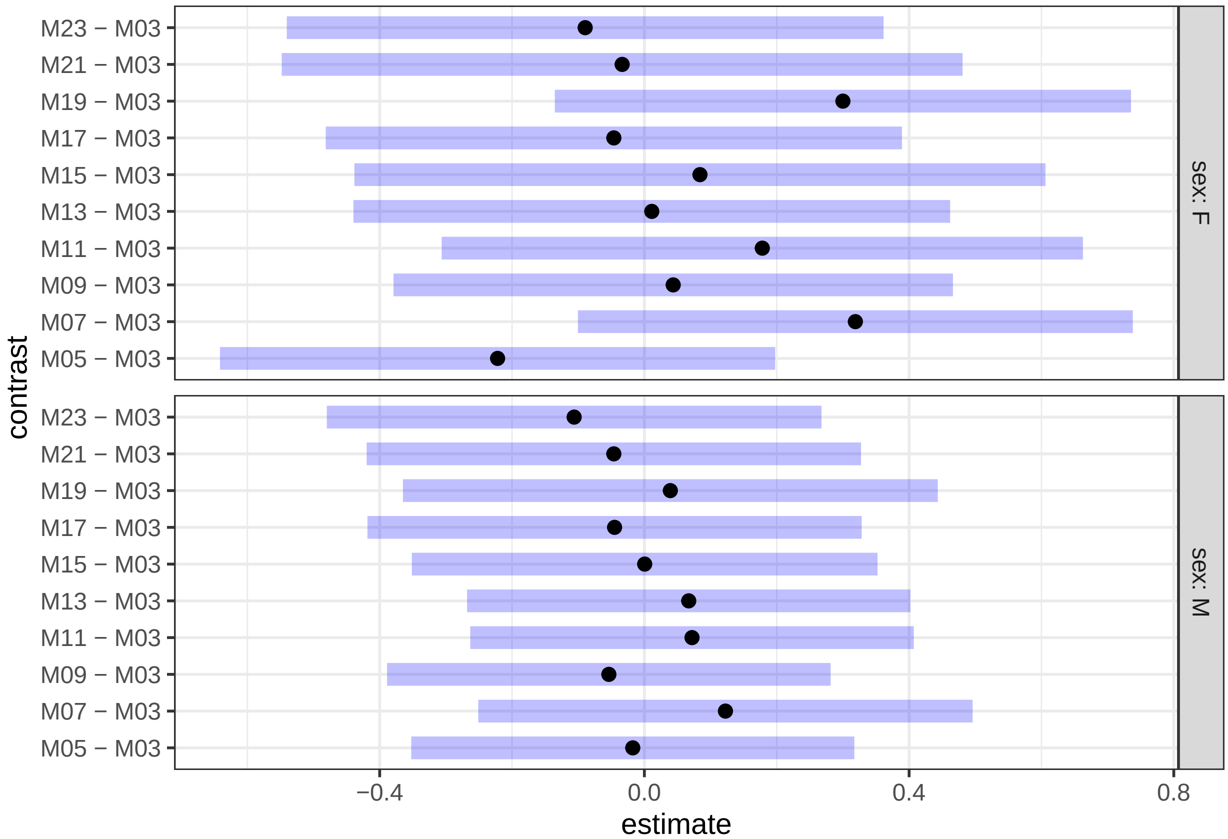
1. No diff at baseline
2. M07 and M19 females are diff from M03
3. Marginal diff between M and F in M19

Table 11: Fixed effects for Observed model (Cohort G2)

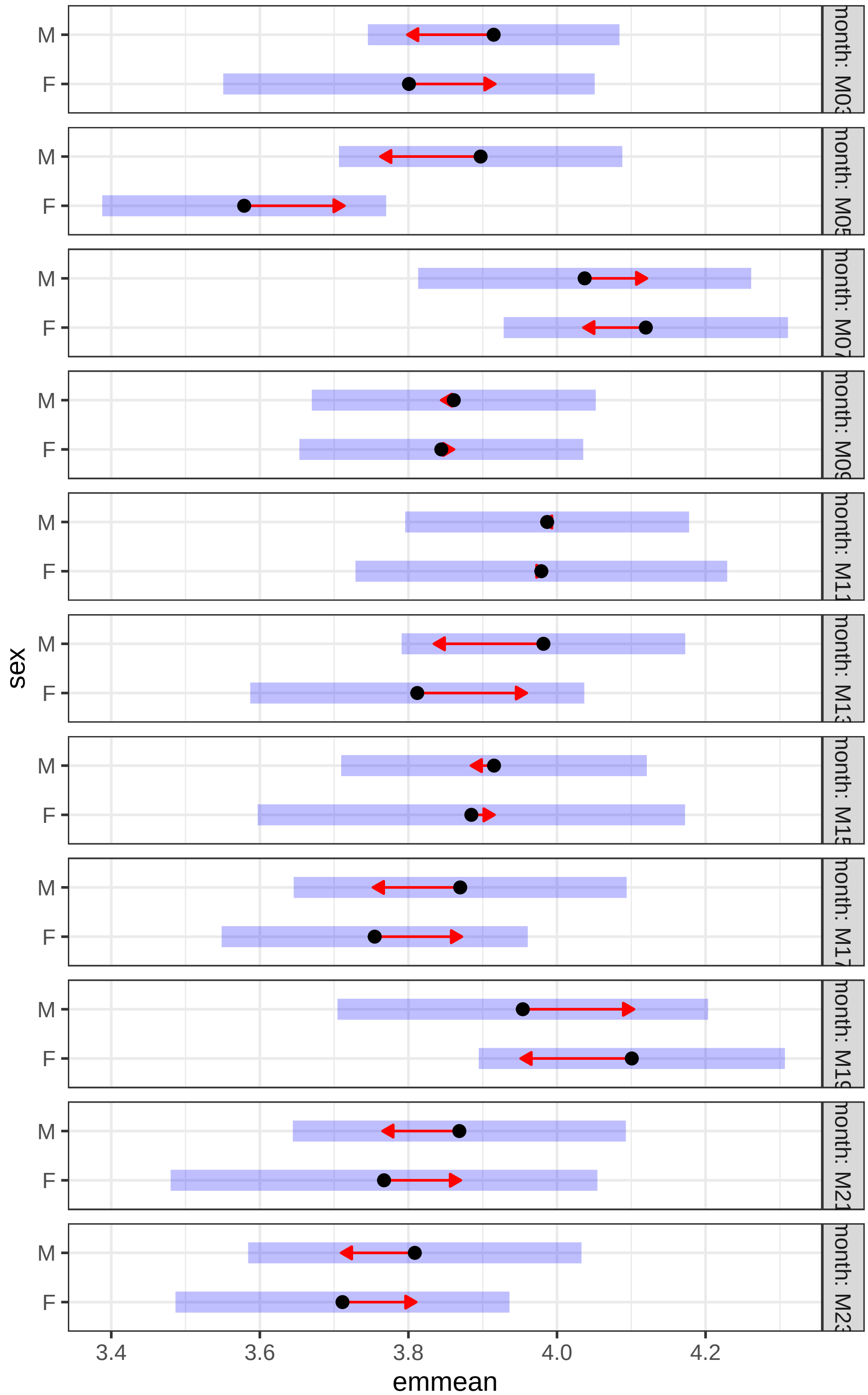
	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	116.9246	11.8680	100.9588	9.8521	0.0000
sexM	13.2977	14.3580	98.5733	0.9261	0.3566
monthM05	11.7280	14.1262	92.5827	0.8302	0.4085
monthM07	45.4176	14.1182	92.3351	3.2170	0.0018
monthM09	24.7788	14.2162	93.7734	1.7430	0.0846
monthM11	13.0188	16.2621	96.9500	0.8006	0.4253
monthM13	3.5086	15.1743	93.7729	0.2312	0.8176
monthM15	16.5746	17.5303	96.4958	0.9455	0.3468
monthM17	17.7518	14.6462	94.5244	1.2120	0.2285
monthM19	56.7518	14.6462	94.5244	3.8748	0.0002
monthM21	-10.5578	17.3059	93.6354	-0.6101	0.5433

	Estimate	Std. Error	df	t value	Pr(>|t|)
monthM23	17.0319	15.1767	93.8676	1.1222	0.2646
sexM:monthM05	15.8732	18.0796	91.9626	0.8780	0.3823
sexM:monthM07	-24.2694	18.9094	92.6070	-1.2835	0.2025
sexM:monthM09	-13.9993	18.1584	92.9841	-0.7710	0.4427
sexM:monthM11	2.7607	19.8010	95.3951	0.1394	0.8894
sexM:monthM13	15.7810	18.9142	92.9558	0.8343	0.4062
sexM:monthM15	-9.2275	21.1623	95.3389	-0.4360	0.6638
sexM:monthM17	-8.5881	19.3048	93.8117	-0.4449	0.6574
sexM:monthM19	-37.3228	19.9866	94.0765	-1.8674	0.0650
sexM:monthM21	5.2741	21.3946	93.3928	0.2465	0.8058
sexM:monthM23	-8.5674	19.7165	93.5724	-0.4345	0.6649

```
# Estimated marginal means by Sex within Month for Shannon
df_wide_G2<- df_wide %>%
  filter(mousegroup.x== "G2")

model1_G2 <- lmer(Shannon ~ sex * month + (1 | mouse), data = df_wide_G2)
emm_sex_within_month_G2 <- emmeans(model1_G2, ~ sex | month)
pairs_sex_within_month_G2 <- contrast(emm_sex_within_month_G2, method = "pairwise", adjust = "holm")

emm_month_within_sex_G2 <- emmeans(model1_G2, ~ month | sex)
baseline_contrasts_G2 <- contrast(emm_month_within_sex_G2, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_G2)
```




```r
plot(emm_sex_within_month_G2, comparisons = TRUE)
```



Cohort G3

OBSERVED

M11 females diff from M03 females

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	151.6000	9.3928	100	16.1401	0.0000
sexM	4.4000	11.5037	100	0.3825	0.7029
monthM05	4.6000	13.2834	100	0.3463	0.7298
monthM07	-2.2667	12.7179	100	-0.1782	0.8589
monthM09	8.4000	15.3383	100	0.5476	0.5852
monthM11	33.9000	14.0892	100	2.4061	0.0180
monthM13	11.1500	14.0892	100	0.7914	0.4306
monthM15	8.0667	15.3383	100	0.5259	0.6001
monthM17	-29.6000	23.0075	100	-1.2865	0.2012
monthM19	-5.8000	13.2834	100	-0.4366	0.6633
monthM22	-20.7667	12.7179	100	-1.6329	0.1056
monthM23	-14.0000	22.0280	100	-0.6356	0.5265
sexM:monthM05	12.5111	16.4187	100	0.7620	0.4479
sexM:monthM07	2.9667	15.8104	100	0.1876	0.8515
sexM:monthM09	-44.2571	18.5039	100	-2.3918	0.0186
sexM:monthM11	-25.0000	16.9331	100	-1.4764	0.1430
sexM:monthM13	13.3500	17.7802	100	0.7508	0.4545
sexM:monthM15	5.3778	18.1215	100	0.2968	0.7673
sexM:monthM17	24.1000	25.4357	100	0.9475	0.3457
sexM:monthM19	-13.6000	17.5723	100	-0.7739	0.4408
sexM:monthM22	13.2667	16.7146	100	0.7937	0.4292

SHANNON

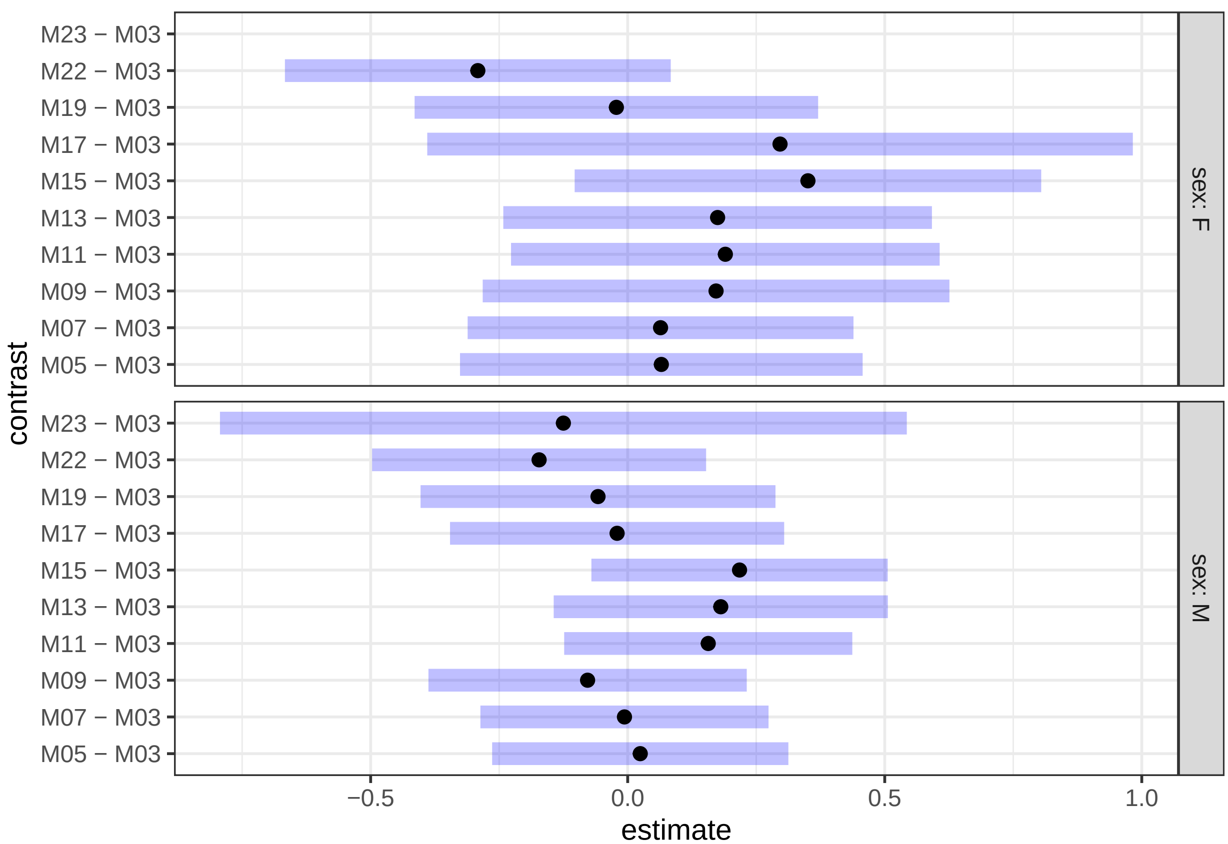
	Estimate	Std. Error	df	t value	Pr()
(Intercept)	3.9141	0.1000	100	39.1539	0.0000
sexM	0.0736	0.1224	100	0.6010	0.5492
monthM05	0.0475	0.1414	100	0.3358	0.7377
monthM07	0.0356	0.1354	100	0.2631	0.7930
monthM09	0.1372	0.1632	100	0.8405	0.4026
monthM11	0.1876	0.1499	100	1.2514	0.2137
monthM13	0.1749	0.1499	100	1.1666	0.2462
monthM15	0.3460	0.1632	100	2.1197	0.0365*
monthM17	0.2787	0.2449	100	1.1383	0.2577
monthM19	-0.0303	0.1414	100	-0.2143	0.8307
monthM22	-0.3130	0.1354	100	-2.3122	0.0228*
sexM:monthM05	-0.0310	0.1747	100	-0.1777	0.8594
sexM:monthM07	-0.0396	0.1683	100	-0.2354	0.8144
sexM:monthM09	-0.2277	0.1969	100	-1.1564	0.2503
sexM:monthM11	-0.0360	0.1802	100	-0.2000	0.8419
sexM:monthM13	0.0120	0.1892	100	0.0635	0.9495
sexM:monthM15	-0.1370	0.1929	100	-0.7106	0.4790
sexM:monthM17	-0.3000	0.2707	100	-1.1080	0.2705
sexM:monthM19	-0.0251	0.1870	100	-0.1342	0.8935
sexM:monthM22	0.1546	0.1779	100	0.8691	0.3869

```r
# Estimated marginal means by Sex within Month for Shannon
df_wide_G3<- df_wide %>%
  filter(mousegroup.x== "G3")

model1_G3 <- lmer(Shannon ~ sex * month + (1 | mouse), data = df_wide_G3)
```

```
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient

## boundary (singular) fit: see help('isSingular')
```

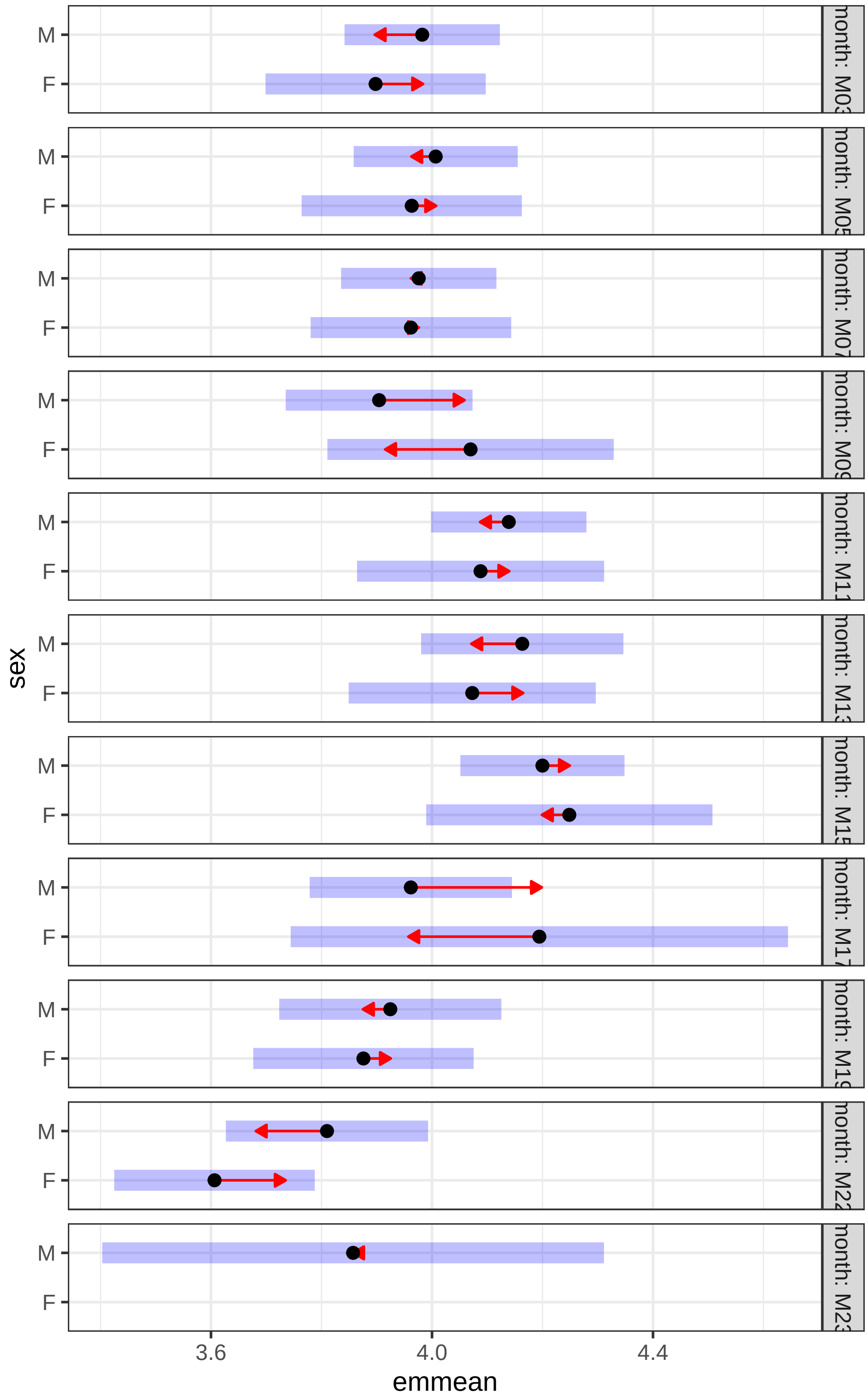
```
emm_sex_within_month_G3 <- emmeans(model1_G3, ~ sex | month)
pairs_sex_within_month_G3 <- contrast(emm_sex_within_month_G3, method = "pairwise", adjust = "holm")

emm_month_within_sex_G3 <- emmeans(model1_G3, ~ month | sex)
baseline_contrasts_G3 <- contrast(emm_month_within_sex_G3, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_G3)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
plot(emm_sex_within_month_G3, comparisons = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_point()‘).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_segment()‘).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_point()‘).
```



Alpha Diversity of mice based on frailty

To determine potential relationship between frailty scores and survival, I filtered the dataset to the last time point of collection

```r
metadata<-read.delim("Young_aging_metadata_v4.txt")

# keeping only the last one per mouse
metadata_sub <- metadata %>%
  group_by(mouse) %>%
  arrange(month_n, .by_group = TRUE) %>%  # ensure chronological order
  slice_tail(n = 1) %>%
  ungroup()
```

This results in 186 mice

```r
#Remove samples with no frailty information
metadata_sub_f<-metadata_sub %>%
  filter(frailty_physical %in% c("PF","NF","F"))

write.table(metadata_sub_f,"metadata_frailty.txt",sep="\t", quote = FALSE,
            row.names = FALSE)

qiime_frail<-qza_to_phyloseq(features="Aging-table-dada2.qza",
                         tree="rooted-tree-merged.qza",
                         metadata="metadata_frailty.txt", taxonomy = "Aging-taxonomy-weighted_silva138-1.

sample_reads <- sample_sums(qiime_frail)
sorted_reads <- sort(sample_reads)
print(sorted_reads)
```

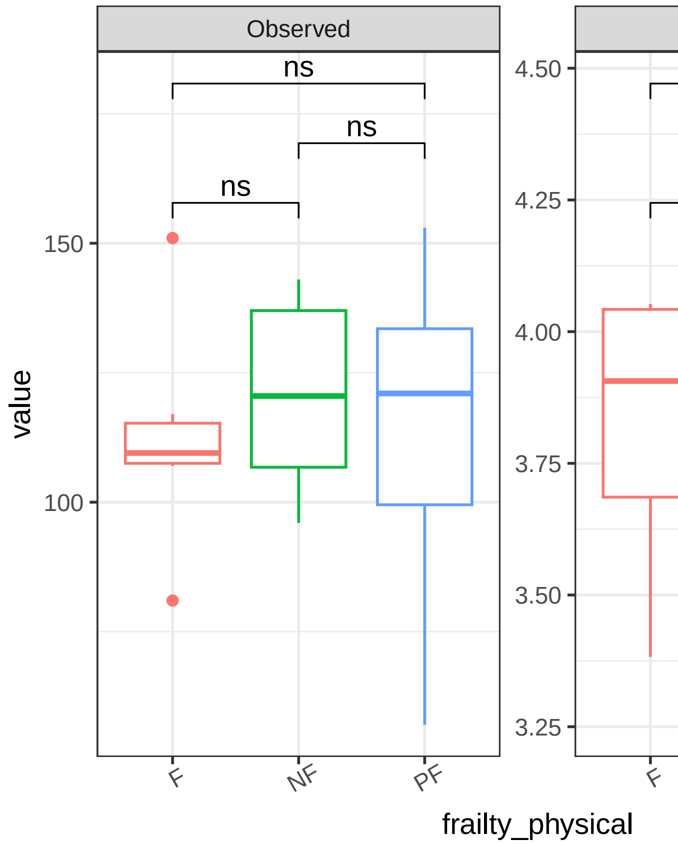
```
## M2201N3M22 F2198N4M22 F2202N1M22 F2202N2M22 M2226N2M17 M2214N3M21 F2199N2M19
##       2235       3122       4328       4359       4762       4831       5191
## F2198N1M22 M2200N3M22 F2198N3M22 M2201N1M22 F2199N3M22 M2201N2M22 M2200N4M22
##       5401       6219       6418       6638       7040       7872       8958
## M2197N2M22 M2212N3M13 M2197N1M22 M2215N2M21 F2211N3M23 F2224N2M22 M2226N3M17
##       8997      10442      10554      10873      13653      16321      16495
## F2223N1M22 F2213N3M23 M2215N1M23 M2214N2M23 M2225N1M22 M2226N4M22 M2226N1M22
##      17012      17486      17781      17844      18262      18311      19661
## M2225N3M22 M2214N1M23 M2225N2M23 F2223N2M22 M2215N3M23 M2225N5M22 F2211N4M23
##      20406      21549      22186      22423      22656      25894      27221
## F2213N1M23 F2224N3M22 M2225N4M22 F2211N1M23 M2212N1M23 F2224N4M22
##      29614      31255      34634      35112      37080      43680
```

```r
qiime.rare.frail<- rarefy_even_depth(qiime_frail,sample.size = min(sample_sums(qiime_frail)))
```

```
## You set `rngseed` to FALSE. Make sure you've set & recorded
##  the random seed of your session for reproducibility.
## See `?set.seed`

## ...

## 3904OTUs were removed because they are no longer
## present in any sample after random subsampling

## ...
```

Alpha diversity

```
metadata_df_frail<-as.data.frame(sample_data( qiime.rare.frail))
metadata_df_frail$SampleID<-rownames(metadata_df_frail)
metadata_df_frail<-as.matrix(metadata_df_frail)

alpha.div.df.frail <- estimate_richness(
  qiime.rare.frail,
  measures=c("Shannon", "Observed")) %>%
  rownames_to_column("SampleID") %>%
  merge(metadata_df_frail, by="SampleID")

alpha.div.df.frail.long <- alpha.div.df.frail %>%
  pivot_longer(c(Shannon, Observed), names_to="metric")

alpha.div.plot.df.frail <- alpha.div.df.frail.long %>%
  arrange(metric, frailty_physical)

a_my_comparisons_frail <-list(c("F","NF"),c("NF","PF"),c("PF","F"))
symnum.args <- list(cutpoints = c(0, 0.0001, 0.001, 0.01, 0.05, 1), symbols = c("****", "***", "**", "*

alpha.div.df.frail.long%>%
  ggplot(aes(x=frailty_physical, y=value, color=frailty_physical))+
geom_boxplot()+facet_wrap(~metric, scales="free")+
  theme_set(theme_bw())+
  theme(axis.text.x=element_text(angle=30))+
  stat_compare_means(comparisons = a_my_comparisons_frail, symnum.args = symnum.args, method = "t.test")
```



I found no association with frailty scores and microbiota

```
#No. sig difference found here
```

```
df_wide_frail<- alpha.div.df.frail.long%>%
  pivot_wider(names_from = metric, values_from = value)
model_shannon_frail <- lmer(Shannon ~ frailty_physical + (1 | mousegroup.x), data = df_wide_frail)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
kable(summary(model_shannon_frail)$coefficients, digits = 4)
```

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.8246	0.0917	38	41.7031	0.0000
frailty_physicalNF	-0.0235	0.1123	38	-0.2091	0.8355
frailty_physicalPF	-0.0519	0.1030	38	-0.5035	0.6175

```
df_wide_frail$howlette_frailty<-as.numeric(df_wide_frail$howlette_frailty)
model_shannon_hf <- lmer(Shannon ~ howlette_frailty + (1 | mousegroup.x), data = df_wide_frail)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
kable(summary(model_shannon_hf)$coefficients, digits = 4)
```

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.8129	0.1026	39	37.1575	0.000
howlette_frailty	-0.1015	0.4040	39	-0.2512	0.803